10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 11.2 — Information Handling (Measures of Dispersion)

Unit 11 | Exercise 11.2 | New Book (PECTAA)

Exercise 11.2 (Solved)

This exercise uses numbers only: range, variance and standard deviation of ungrouped and grouped data. In every question the table of working is the answer, so write the table first and then the formulae.

1. Find the range of the following data sets: (i) 63, 89, 98, 125, 79, 108, 117, 60 (ii) 43.5, 13.6, 18.9, 38.4, 61.4, 29.4

(i) Highest value H = 125, lowest value L = 60

Range = H − L = 125 − 60 = 65

(ii) Highest value H = 61.4, lowest value L = 13.6

Range = H − L = 61.4 − 13.6 = 47.8

Answer: (i) Range = 65 (ii) Range = 47.8

2. If the range and the lowest value of a set of data are 46.7 and 13.4 respectively, then find the highest value.

Range = highest value − lowest value

46.7 = H − 13.4

H = 46.7 + 13.4 = 60.1

Answer: Highest value = 60.1

3. Calculate the range of the following data:

Income of workers

Income (in Rs.)	4000 – 4500	4500 – 5000	5000 – 5500	5500 – 6000	6000 – 6500
No. of workers	8	12	30	21	6

The range of grouped data is found from the class limits.

Highest value H = 6500, lowest value L = 4000

Range = H − L = 6500 − 4000 = 2500

Answer: Range = Rs. 2500

4. A group of 7 workers reported the number of items they assembled in a day as: 52, 55, 50, 53, 54, 56, 52. Find the standard deviation and variance of the items assembled.

Here n = 7.

Mean = (52 + 55 + 50 + 53 + 54 + 56 + 52) / 7 = 372 / 7 = 53.14

Table for variance ($\bar{x}$ = 53.14)

x	x − $\bar{x}$	(x − $\bar{x}$)²
52	−1.14	1.30

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55	1.86	3.46
50	−3.14	9.86
53	−0.14	0.02
54	0.86	0.74
56	2.86	8.18
52	−1.14	1.30
Σ	0	Σ(x − x̄)² = 24.86

$\sigma^2 = \Sigma(x - \bar{x})^2 / n = 24.86 / 7 = 3.55$

$\sigma = \sqrt{3.55} = 1.88$

Answer: Variance = 3.55, standard deviation = 1.88 items

5. A librarian recorded the number of visitors during 5 days of a week: 120, 135, 130, 125, 140. Calculate the variance and standard deviation of visitors.

Here n = 5.

Mean = (120 + 135 + 130 + 125 + 140) / 5 = 650 / 5 = 130

Table for variance ($\bar{x}$ = 130)

x	x − x̄	(x − x̄)²
120	−10	100
135	5	25
130	0	0
125	−5	25
140	10	100
Σ	0	250

$\sigma^2 = \Sigma(x - \bar{x})^2 / n = 250 / 5 = 50$

$\sigma = \sqrt{50} = 7.07$

Answer: Variance = 50, standard deviation = 7.07 visitors

6. Find the range, variance and standard deviation of first 23 odd numbers.

The first 23 odd numbers are 1, 3, 5, …, 45, so n = 23.

Range = H − L = 45 − 1 = 44

Mean = (1 + 3 + 5 + … + 45) / 23 = 529 / 23 = 23

For odd numbers the mean is the middle number, which is 23.

Now $x - \bar{x} = \pm 2, \pm 4, \ldots, \pm 22$ (each value twice except 0 at the centre).

$\Sigma(x - \bar{x})^2 = 2(2^2 + 4^2 + 6^2 + \ldots + 22^2) = 2(4 + 16 + 36 + 64 + 100 + 144 + 196 + 256 + 324 + 400 + 484)$

$\Sigma(x - \bar{x})^2 = 2(2024) = 4048$

$\sigma^2 = \Sigma(x - \bar{x})^2 / n = 4048 / 23 = 176$

$\sigma = \sqrt{176} = 13.27$

Answer: Range = 44, variance = 176, standard deviation = 13.27

7. The rainfall recorded in various places of five districts in a week is given below. Find its variance and standard deviation.

Rainfall data

Rainfall (in mm)	42	51	54	61	63	71
Number of places	5	13	4	9	5	4

Here $\Sigma f = 5 + 13 + 4 + 9 + 5 + 4 = 40$

$\Sigma fx = 5(42) + 13(51) + 4(54) + 9(61) + 5(63) + 4(71) = 210 + 663 + 216 + 549 + 315 + 284 = 2237$

Mean $= \Sigma fx / \Sigma f = 2237 / 40 = 55.93$

Table for variance ($\bar{x} = 55.93$)

x	f	fx	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
42	5	210	−13.93	194.04	970.20
51	13	663	−4.93	24.30	315.90
54	4	216	−1.93	3.72	14.88
61	9	549	5.07	25.70	231.30
63	5	315	7.07	49.98	249.90
71	4	284	15.07	227.10	908.40
Σ	40	2237	0		2690.60

$\sigma^2 = \Sigma f(x - \bar{x})^2 / \Sigma f = 2690.60 / 40 = 67.27$

$\sigma = \sqrt{67.27} = 8.20$

Answer: Variance = 67.27 (mm²), standard deviation = 8.20 mm

8. Machine A output (units): 98, 100, 102, 101, 99. Machine B output (units): 95, 100, 105, 90, 110. (i) Which machine has better performance? (ii) Which machine is more consistent?

Machine A: n = 5

Mean $= (98 + 100 + 102 + 101 + 99) / 5 = 500 / 5 = 100$

Working for Machine A ($\bar{x} = 100$)

x	$x - \bar{x}$	$(x - \bar{x})^2$
98	−2	4
100	0	0
102	2	4
101	1	1
99	−1	1
Σ	0	10

$\sigma^2(A) = 10 / 5 = 2$, $\sigma(A) = \sqrt{2} = 1.41$ units

Machine B: n = 5

Mean = (95 + 100 + 105 + 90 + 110) / 5 = 500 / 5 = 100

Working for Machine B ($\bar{x}$ = 100)

x	$x - \bar{x}$	$(x - \bar{x})^2$
95	−5	25
100	0	0
105	5	25
90	−10	100
110	10	100
Σ	0	250

$\sigma^2(B) = 250 / 5 = 50$, $\sigma(B) = \sqrt{50} = 7.07$ units

Both means are equal, so both machines give the same average production. The standard deviation of A is much smaller, so its output is steadier.

Answer: (i) Both perform equally well (mean = 100 units) (ii) Machine A is more consistent because σ(A) = 1.41 < σ(B) = 7.07

9. The monthly sales (rupees in lakhs) for two salespersons over 6 months are — Person A: 5.5, 5.7, 5.4, 5.6, 5.8, 5.6; Person B: 5.5, 6.5, 4.5, 6.0, 5.0, 6.0. Compare their performance and consistency.

Person A: n = 6

$\Sigma x = 5.5 + 5.7 + 5.4 + 5.6 + 5.8 + 5.6 = 33.6$

Mean = 33.6 / 6 = 5.6

Deviations: −0.1, 0.1, −0.2, 0, 0.2, 0 → squares 0.01, 0.01, 0.04, 0, 0.04, 0

$\Sigma(x - \bar{x})^2 = 0.10$, $\sigma^2(A) = 0.10 / 6 = 0.0167$

$\sigma(A) = \sqrt{0.0167} = 0.13$ lakh

Person B: n = 6

$\Sigma x = 5.5 + 6.5 + 4.5 + 6.0 + 5.0 + 6.0 = 33.5$

Mean = 33.5 / 6 = 5.58

Deviations: −0.08, 0.92, −1.08, 0.42, −0.58, 0.42

$\Sigma(x - \bar{x})^2 = 0.0064 + 0.8464 + 1.1664 + 0.1764 + 0.3364 + 0.1764 = 2.7084$

$\sigma^2(B) = 2.7084 / 6 = 0.4514$

$\sigma(B) = \sqrt{0.4514} = 0.67$ lakh

The means differ by only Rs. 2000, so their average performance is almost the same. The sales of A stay very close to the mean, while the sales of B jump about.

Answer: Both perform almost equally (means 5.6 and 5.58 lakh); Person A is far more consistent because σ(A) = 0.13 < σ(B) = 0.67

10. The table given below shows the daily wages of workers in a textile mill, grouped into six income brackets. Calculate the mean, variance and standard deviation of the wages.

Daily wages of workers

Daily wage (Rs.)	800 – 1000	1000 – 1200	1200 – 1400	1400 – 1600	1600 – 1800	1800 – 2000
Frequency	2	4	6	8	2	1

We take the class marks x as the mid-values of the classes.

Table for mean and variance

Class	f	x	fx	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
800 – 1000	2	900	1800	−460.87	212401	424802
1000 – 1200	4	1100	4400	−260.87	68053	272212
1200 – 1400	6	1300	7800	−60.87	3705	22230
1400 – 1600	8	1500	12000	139.13	19357	154856
1600 – 1800	2	1700	3400	339.13	115009	230018
1800 – 2000	1	1900	1900	539.13	290661	290661
Σ	23		31300	0		1394779

Mean = $\Sigma fx / \Sigma f = 31300 / 23 = 1360.87$

$\sigma^2 = \Sigma f(x - \bar{x})^2 / \Sigma f = 1394779 / 23 = 60642.57$

$\sigma = \sqrt{60642.57} = 246.26$

Answer: Mean = Rs. 1360.87, variance = 60642.57, standard deviation = Rs. 246.26

11. A company forecasts monthly sales (rupees in millions): 15, 18, 14, 20, 13. Find variability in sales predictions.

Here n = 5.

Mean = (15 + 18 + 14 + 20 + 13) / 5 = 80 / 5 = 16

Deviations: −1, 2, −2, 4, −3 → squares 1, 4, 4, 16, 9

$\Sigma(x - \bar{x})^2 = 34$

$\sigma^2 = 34 / 5 = 6.8$, $\sigma = \sqrt{6.8} = 2.61$ million

Answer: Variance = 6.8, standard deviation = 2.61 million rupees

12. Unemployment rates (%) in five provinces are 5.2, 6.0, 4.8, 5.5, 6.2. Calculate standard deviation and describe is there balanced unemployment rate?

Here n = 5.

Mean = (5.2 + 6.0 + 4.8 + 5.5 + 6.2) / 5 = 27.7 / 5 = 5.54

Table for standard deviation ($\bar{x} = 5.54$)

x	$x - \bar{x}$	$(x - \bar{x})^2$
5.2	−0.34	0.1156
6.0	0.46	0.2116
4.8	−0.74	0.5476
5.5	−0.04	0.0016
6.2	0.66	0.4356

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Σ	0	1.3120

$\sigma^2 = 1.3120 / 5 = 0.2624$

$\sigma = \sqrt{0.2624} = 0.51$ %

The standard deviation is very small, so the rates of all five provinces stay close to the mean of 5.54%.

Answer: σ = 0.51%; yes, the unemployment rate is almost balanced in the five provinces

13. Find variance and standard deviation: (i) $\Sigma x = 45$, $\Sigma x^2 = 421$, $n = 5$ (ii) $\Sigma fx = 210$, $\Sigma fx^2 = 7560$, $\Sigma f = 6$ (iii) $\bar{x} = 18$, $\Sigma fx^2 = 1670$, $\Sigma f = 5$

(i) $\sigma^2 = \Sigma x^2 / n - (\Sigma x / n)^2$

$\sigma^2 = 421 / 5 - (45 / 5)^2 = 84.2 - 81 = 3.2$

$\sigma = \sqrt{3.2} = 1.79$

(ii) $\sigma^2 = \Sigma fx^2 / \Sigma f - (\Sigma fx / \Sigma f)^2$

$\sigma^2 = 7560 / 6 - (210 / 6)^2 = 1260 - 1225 = 35$

$\sigma = \sqrt{35} = 5.92$

(iii) $\sigma^2 = \Sigma fx^2 / \Sigma f - (\bar{x})^2$

$\sigma^2 = 1670 / 5 - (18)^2 = 334 - 324 = 10$

$\sigma = \sqrt{10} = 3.16$

Answer: (i) $\sigma^2 = 3.2$, $\sigma = 1.79$ (ii) $\sigma^2 = 35$, $\sigma = 5.92$ (iii) $\sigma^2 = 10$, $\sigma = 3.16$

Remember

Range = H − L, the quickest measure of spread.

$\sigma^2 = \Sigma(x - \bar{x})^2 / n$ (ungrouped) and $\Sigma f(x - \bar{x})^2 / \Sigma f$ (grouped).

σ is always positive, and it is in the same unit as the data.

Smaller σ means more consistent or more stable data.